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### REPORTING CHANNEL STATE INFORMATION IN A MULTIPLE COMPONENT CARRIER NETWORK

#### Abstract

A user equipment (UE) may determine channel state information (CSI) reports with each CSI report being related to a component carrier. The UE may send a subframe with CSI report(s) on a physical uplink control channel (PUCCH). When a collision occurs in the subframe having the CSI report(s) on the PUCCH a lower priority CSI report may be dropped from the transmission.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/385,431, filed Jul. 26, 2021, which is a continuation of U.S. patent application Ser. No. 16/780,452, filed Feb. 3, 2020, which issued as U.S. Pat. No. 11,076,312 on Jul. 27, 2021 which is a continuation of U.S. patent application Ser. No. 15/483,699, filed Apr. 10, 2017, which issued as U.S. Pat. No. 10,555,198 on Feb. 4, 2020, which is a continuation of U.S. patent application Ser. No. 15/137,940 filed Apr. 25, 2016, which issued as U.S. Pat. No. 9,622,227 on Apr. 11, 2017, which is a continuation of U.S. patent application Ser. No. 13/948,852 filed Jul. 23, 2013, which issued as U.S. Pat. No. 9,325,479 on Apr. 26, 2016, which is a continuation of U.S. patent application Ser. No. 12/752,572 filed Apr. 1, 2010, which issued as U.S. Pat. No. 8,520,491 on Aug. 27, 2013, all of which are incorporated by reference as if fully set forth.

## FIELD OF INVENTION

[0002] The present invention relates generally to communications and particular to communications over multiple carriers. More particularly, in certain embodiments the present invention provides a mechanism for periodic channel state information signaling that employ carrier aggregation.

## BACKGROUND

[0003] There has not been any significant prior work on related signaling details of periodic channel state information (CSI) with multiple parallel configurations in the case of multiple component carriers (CCs). Contributions in the Third Generation Partnership Project (3GPP) tend to focus on extending the physical uplink control channel (PUCCH) payload beyond 11 bits, which would permit new, larger CSI formats.

[0004] Activation and deactivation of component carriers has been discussed. There has also been some discussion about the concept of primary and secondary component carriers in radio access network 2 (RAN2): primary component carrier (PCC), and secondary component carrier (SCC), respectively.

## SUMMARY

[0005] In one embodiment, the present invention is a method. The method includes determining, in an apparatus, channel state information. The channel state information includes channel state

information of a plurality of component carriers. The method also includes reporting, by the apparatus, the channel state information comprising a plurality of reports in an aggregated form. [0006] In a further embodiment, the present invention is an apparatus. The apparatus includes at least one memory including computer program code. The apparatus also includes at least one processor. The at least one memory and the computer program code are configured to, with the at least one processor, cause the apparatus at least to determine channel state information. The channel state information includes channel state information of a plurality of component carriers. The at least one memory and the computer program code are also configured to, with the at least one processor, cause the apparatus at least to report the channel state information comprising a plurality of reports in an aggregated form.

[0007] In a further embodiment, the present invention is a computer-readable non-transitory medium encoded with information that, when executed in hardware, performs a process. The process includes determining channel state information. The channel state information includes channel state information of a plurality of component carriers. The process also includes reporting the channel state information comprising a plurality of reports in an aggregated form.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0008] For proper understanding of the invention, reference should be made to the accompanying drawings, wherein:

[0009] FIG. 1 illustrates an example of an implementation for handling collisions between channel state information reports.

[0010] FIG. 2 illustrates a method according to certain embodiments of the present invention.

[0011] FIG. 3 illustrates an apparatus according to certain embodiments of the present invention.

[0012] FIG. 4 illustrates a technique for constructing rules for handling component carrier aggregation.

### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

[0013] Channel State Information (CSI) feedback signaling with carrier aggregation may be useful in the Long Term Evolution (Advanced) (LTE-Advanced) of the Third Generation Partnership Project (3GPP) system, which may be a part of 3GPP LTE Release 10 (3GPP LTE Rel-10).

[0014] LTE-Advanced may serve as an evolution of the LTE Release 8 (Rel-8) system, fulfilling the International Telecommunication Union (ITU) Radiocommunication Sector (ITU-R) requirements for International Mobile Telecommunications-Advanced (IMT-Advanced). Carrier aggregation will be one technology component aimed at providing high bandwidth and peak data rates for the new system.

[0015] An efficient way to provide channel state information (CSI) reporting for multiple component carriers (CCs) can help to make efficient use of aggregated spectrum. Periodic CSI signaling aspects on the physical uplink control channel (PUCCH) and especially the rules and procedures related to handling multiple CSI reporting configurations will be discussed below.

[0016] Periodic reporting of CSI such as channel quality indicator (CQI), precoding matrix indicator (PMI), and rank indicator (RI) is the fundamental mode of feedback signaling in LTE Rel-8. The periodic CSI reports are carried on PUCCH, and the payload size is typically limited to a maximum of 11 bits. Due to the limited payload size, the reports typically contain little or no information about the frequency domain behavior of the channel.

[0017] With carrier aggregation, the need for periodic CSI may remain. LTE-Advanced (Rel-10) may, for example, support approximately five downlink (DL) CCs. Hence the straightforward expansion of Rel-8 reporting for multiple CCs would result in large reports (e.g.  $5 \times 11$  bits = 55 bits). This may not be helpful from the uplink (UL) signaling point of view. First, such high overhead

may limit uplink capacity considerably. Second, in many cases it may not be possible to guarantee sufficient UL coverage for such large payloads. Hence some compression methods may be used to reduce the UL signaling burden.

[0018] The most straightforward way to reduce the signaling load is to configure CSI reporting separately for each CC. This option can assist standardization, maximizing the commonality with Rel-8 specifications. Furthermore, the DL transmission mode can be configured separately for each DL CC, which would make joint encoding of the CSI reports complicated, or at least lead to a large number of supported options.

[0019] There are a number of ways that component carriers can be handled in a communication system. For example, there can be explicit activation and deactivation of configured downlink (DL) component carriers is done by media access control (MAC) signaling. There can also be implicit deactivation of configured DL component carriers. The configuration of a DL/UL component carrier can be within a user equipment's (UE's) aggregation capability. As a consequence, activation/deactivation as well can be within UE capability.

[0020] A newly configured component carrier can always be in a default state of “deactivated.” In order to activate a newly configured component carrier, an activation command may be necessary. Furthermore, DL component carriers can be activated and deactivated individually. A single activation/deactivation command can activate/deactivate a subset of the configured DL component carriers.

[0021] The uplink (UL) primary component carrier (PCC) and DL PCC can be configured per UE. The UL PCC can be used for transmission of L1 uplink control information. In certain embodiments, the DL PCC cannot be de-activated. Re-establishment can be triggered when the DL PCC experiences radio link failure (RLF), as opposed to when other DL CCs experience RLF. These implementations, however, are merely examples of how component carriers can be handled.

[0022] The following discussion focuses on the rules and procedures related to handling multiple CSI reporting configurations. Such rules for handling multiple CSI reporting configurations may be useful when transmitting a CSI for multiple CCs simultaneously in the same subframe. This situation may be described as being the state that exists when multiple configurations collide.

[0023] The rules and procedures may relate to handling periodic CSI reporting for carrier aggregation using PUCCH with multiple parallel configurations. The multiple parallel configurations may be CC-specific.

[0024] For each configured DL CC, a periodic CSI reporting mode can be configured. Reporting disabled is here considered as a reporting mode, a special case of configuration. In other words, the reporting disabled case can be considered a periodic report having the frequency of “never.” The configuration can be done using RRC signaling and includes parameters such as periodicity, subframe offset, reporting mode, PUCCH resource used for reporting etc.

[0025] For the primary component carrier (PCC) the UE can start reporting CQI after it has received the CQ configuration via radio resource control (RRC) signaling. This follows similar procedures as in LTE Rel-8.

[0026] For the secondary component carriers (SCC) the UE can start to report CQI according to the RRC configuration after it has received an activation command for the given CC with media access control (MAC) signaling.

[0027] In the case of CC deactivation, the UE can stop reporting CSI for that CC. In other words, there may be no requirement that the UE report regarding inactive CCs.

[0028] In certain embodiments, when the periodic reports for two or more CCs would collide, the following complementary rules take place. The rules are numbered for the sake of identification. Unless otherwise noted, one should not treat the numbers as though they were setting forth a fixed sequence, such that, for example, 2 must come after 1 and before 3.

[0029] Option 1 is multiplexing. In this option, when the PUCCH resources for the colliding CSI reports are located on the same or the adjacent PRB, all the CSI reports can be sent simultaneously.

Each UE may be configured to omit this option and always perform the dropping of colliding CSI reports according to the following options, which involve prioritization.

[0030] Option 2 gives infrequent reports a priority. In this option, the CC configuration with lowest periodicity is prioritized and the other simultaneous reports are omitted. In the case of equal periodicity, a CSI corresponding to a pre-determined CC is sent while the CSI(s) corresponding to other CC(s) is/are dropped, similar to Option 4 discussed below.

[0031] Option 3 gives rank indicators priority. In this option, a report containing RI is prioritized over CQI/PMI. In the case when the need to transmit them simultaneously occurs, CQI/PMI is dropped.

[0032] Option 4 ranks different CCs separately. The priority order for different CCs can be configured separately. This can be done with e.g. RRC configuration. The configuration indicates in which order the CSI from different CCs is prioritized. In the case of collision the CSI for the CC with higher priority is sent while the other one is dropped.

[0033] Option 5 gives priority to primary component carriers over secondary component carriers. In this option, CSI reports for PCC are prioritized over SCC.

[0034] Various combinations of different options are also possible. For example, Option 2 and Option 3 can be combined. Option 1 can be also combined with Option 2, 3, 4 or 5 in the case that some but not all CSI reports are located on the same or the adjacent PRB. Thus, for example, a first single CSI report according to one of options 2-5 is selected, and after that other CSI reports on same or adjacent PRB are also selected for transmission.

[0035] The reporting configuration for the DL PCC can follow the Rel-8 principles. For the SCC(s) the CSI reporting can be masked with the activation/deactivation command received via MAC signaling. There can be, for example, a bitmap indicating which of the SCCs are activated or deactivated.

[0036] An example of an implementation for handling collisions between CSI reports can be summarized with the block diagram shown in FIG. 1. The embodiment of FIG. 1 shows a decision tree, which can be one way in which certain embodiments of the present invention is implemented.

[0037] As shown in FIG. 1, when the need arises to transmit CSI reports for two or more CCs simultaneously occurs (first yes branch from the top of the diagram), a device can first check whether the PUCCH resources configured for transmission of the CSI reports are on a single, or alternatively on adjacent, PRBs. One reason for imposing such a restriction is that transmission on multiple non-adjacent PUCCHs could violate single-carrier properties, resulting in increased cubic metric and out-of-band emissions.

[0038] If the colliding reports meet the criteria, as illustrated, for example, in the second yes branch from the top of the diagram, the reports are sent according to their respective configurations. If the colliding reports do not meet the criteria, as illustrated for example, in the lower no branch, only the reports with highest priority are sent. To determine the highest priority, options 2, 3, 4, or 5, discussed above or a combination thereof can be employed.

[0039] Alternatively, the multiplexing functionality may also be configurable. For example, one can limit a cell edge user equipment (UE) not to send more than one CSI report at a time in any case.

[0040] The condition for concurrent transmission of multiple PUCCH Format 2/2a/2b resources can be formulated in the following way (N<sub>x</sub>PUCCH Format 2/2a/2b):

[0041] First, a physical resource block (PRB) index  $m$  can be defined for a  $k$ th PUCCH Format 2/2a/2b resource as

$$[00001] m(k) = \text{floor} \left( \frac{n_{\text{PUCCH}}^{(2)}(k)}{N_{\text{sc}}^{\text{RB}}} \right)$$

where  $n_{\text{PUCCH}}^{(2)}$  is the higher-layer configured resource index for a periodic CSI transmitted on PUCCH and  $N_{\text{sc}}^{\text{RB}}$  is the number of sub-carriers (SC) per PRB (there may be 12 sub-carriers per resource block), respectively.

[0042] The criteria for concurrent transmission of multiple PUCCH format 2/2a/2b resources can

be formulated as

[00002]  $m(k) \in [b, (b + 2n)]$ ,

where  $b$  is an integer and  $n \in [0, 1, \dots, N]$  where  $N$  is a predefined integer. Integer  $b$  can be defined as  $b = \min(m(k))$ . This criteria ensures that PUCCH resources transmitted at the same time are located on the same side of the system bandwidth. There are two special cases with  $n=0$  and  $n=1$ : [0043]  $n=0$ : PUCCH resources are located on the same PRB; and [0044]  $n=1$ : PUCCH resources are located on adjacent PRBs.

[0045] In principle, PUCCH resources can be located on multiple adjacent PRBs. To also cover this case, the criteria can be checked sequentially: sort  $m(k)$  in increasing order,  $N=1$ ,  $b = \min(m(k))$ ; and starting from smallest  $m(k)$ , check criteria. If it is filled, update  $N = m(k) + 1$  for current  $k$  and continue.

[0046] Certain embodiments of the present invention may have various advantages. For example, certain embodiments of the present invention may permit independent configuration per CC, which may permit reuse of the Rel-8 functionality as much as possible. Additionally, since there are no existing solutions for handling collisions between CSI reports, the approach above may be implemented widely, such as by means of a standard. Moreover, allowing for multiplexing of CSIs when they are transmitted on a single or adjacent PRBs brings advantages in terms of performance, assuming that sufficient coverage can be guaranteed.

[0047] FIG. 2 illustrates a method according to certain embodiments of the present invention. The method can be performed by an apparatus such as a terminal, a user equipment, or a mobile node. The method of FIG. 2 includes determining **210**, in an apparatus, channel state information. The channel state information includes channel state information of a plurality of component carriers. The method also includes periodically reporting **220**, by the apparatus, the channel state information comprising a plurality of reports in an aggregated form. An aggregated form can include several independent reports. The periodically reporting **220** can be performed via a physical uplink control channel.

[0048] The method can include preparing **215** the aggregated form. The preparing **215** can include dropping **217** at least the channel state information report for at least one configuration based on a priority. It should be noted that a configuration for a component carrier may include more than one report.

[0049] After channel state information of a first component carrier is selected for inclusion in the aggregated form based on a priority, other channel state information corresponding to component carriers that have the same or an adjacent physical resource block can also be selected for inclusion in the aggregated form. The priority can be, for example, lowest periodicity of reporting. In other words, the highest priority can be assigned to a component carrier whose report is to be sent the most infrequently.

[0050] As may be understood from the discussion above, a computer-readable non-transitory medium can be encoded with information that, when executed in hardware, performs a process corresponding to the process disclosed in FIG. 2, or any other process discussed herein. By non-transitory medium, it is meant that the medium is not a transitory signal. Examples of non-transitory mediums include a computer-readable medium, a computer distribution medium, a computer-readable storage medium, and a computer program product.

[0051] Channel state information for component carrier configurations having equal periodicity can be prioritized according to a predetermined ranking for component carriers. Channel state information for component carrier configurations comprising a rank indicator can be prioritized above channel state information for component carrier configurations comprising channel quality indicator or precoding matrix indicator. Alternatively, or in addition, a plurality of component carrier configurations can be individually assigned a predetermined priority. Channel state information for primary component carriers can be prioritized over channel state information for secondary component carriers.

[0052] FIG. 3 illustrates an apparatus according to certain embodiments of the present invention. The apparatus can include at least one memory **310** including computer program code **320**. The apparatus can also include at least one processor **330**. The apparatus can be configured to communicate using a transceiver **340**, including a receiver portion **343** and a transmitter portion **347**, with a base station or enhanced node B (not illustrated). The apparatus can use an antenna **350** to communicate with the base station or other network element over a wireless link **360**, which may be a cellular radio link. The apparatus can be configured to prepare reports regarding, for example, downlink (DL) quality for component carriers of the wireless link **360**.

[0053] In the process, the at least one memory **310** and the computer program code **320** can be configured to, with the at least one processor **330**, cause the apparatus at least to determine channel state information. The channel state information can be channel state information of a plurality of component carriers. The at least one memory **310** and the computer program code **320** can also be configured to, with the at least one processor **330**, cause the apparatus at least to periodically report the channel state information comprising a plurality of reports in an aggregated form.

[0054] The at least one memory **310** may be any suitable form of memory, such as the onboard memory of a controller chip, a hard drive, various types of Random Access Memory (RAM), or Read-Only Memory (ROM). The computer program code **320** can be any appropriate form of computer instructions. The computer program code **320** may be provided in a compiled or interpreted form. The processor **330** can be any suitable processing device, such as but not limited to a Central Processing Unit (CPU), controller, or Application Specific Integrated Circuit (ASIC).

[0055] FIG. 4 illustrates a technique for constructing rules for handling component carrier aggregation. As illustrated in FIG. 4, the technique can begin by selecting **410** a prioritization technique from a variety of techniques. Four such techniques are illustrated, although other techniques are permitted.

[0056] The first of the four techniques shown is to prioritize **420** by frequency/periodicity of reports. That is to say, a CC report that is to be infrequently reported may be given a higher priority than frequently reported CC reports or vice versa.

[0057] The second of the four techniques shown is to prioritize **430** by preferring RI over CGI/PMI (or vice versa). This technique permits the most important aspects of a CC report to be given priority over those that the system considers less important.

[0058] To prioritize **440** by ranking CCs in separate, predetermined ways is the third of the four techniques shown in FIG. 4. In this technique, each individual CC may be assigned a predetermined ranking and may be prioritized according to the predetermined ranking.

[0059] The fourth technique of the four techniques in FIG. 4 is to prioritize **450** by preferring CSI reports for PCCs over those for SCCs. This technique can permit the reports for PCCs to be given a preference, when it is not feasible to report on both PCCs and SCCs. Like the other priority schemes it can be reversed, if desired, such that SCCs are given a priority over PCCs.

[0060] Finally, in addition to the selected CC report of the various available CC reports, the system can multiplex **460** any “close” reports. In this case, “close” can refer to reports for the same or an adjacent physical resource block (PRB).

[0061] One having ordinary skill in the art will readily understand that the invention as discussed above may be practiced with steps in a different order, and/or with hardware elements in configurations which are different than those which are disclosed. Therefore, although the invention has been described based upon these preferred embodiments, it would be apparent to those of skill in the art that certain modifications, variations, and alternative constructions would be apparent, while remaining within the spirit and scope of the invention. For example, although certain embodiments of the present invention have been discussed in relation to 3GPP system, it should be understood that the present invention may be applied to other system such as the Evolved Universal Terrestrial Radio Network (E-UTRAN) and the like. In order to determine the metes and bounds of the invention, therefore, reference should be made to the appended claims.

## Claims

1. A method for use in a base station the method comprising: receiving, from a user equipment (UE), when two channel state information (CSI) reports are to be transmitted by the UE on a physical uplink channel in a same subframe, and each of the two of CSI reports relate to a respective one of a plurality of component carriers, and one of the two CSI reports is a CSI report containing a Rank Indicator (RI), and one of the two CSI reports is a CSI report containing a Channel Quality Indicator (CQ), only the CSI report containing the RI on the physical uplink channel; and receiving, from the UE, when two CSI reports are to be transmitted by the UE on a physical uplink channel in a same subframe, and each of the two of CSI reports relate to a respective one of a plurality of component carriers, and each of the two CSI reports is a CSI report containing a Rank Indicator (RI), only the CSI report containing a RI associated with a highest priority component carrier.
2. The method of claim 1, wherein the physical uplink channel is a physical uplink control channel (PUCCH).
3. The method of claim 1, wherein a CSI report is received from the UE every subframe.
4. The method of claim 1, further comprising: transmitting, to the UE, information to configure the UE with component carrier priority information.
5. The method of claim 1, further comprising: transmitting, to the UE, information to configure the UE to perform periodic CSI reporting associated with one or more of the plurality of component carriers.
6. The method of claim 1, further comprising: transmitting, to the UE, a medium access control (MAC) activation command to activate one of the plurality of component carriers.
7. A base station comprising: a baseband processor; and a radio frequency (RF) transceiver, the baseband processor and RF transceiver operable to receive, from a user equipment (UE), when two channel state information (CSI) reports are to be transmitted by the UE on a physical uplink channel in a same subframe, and each of the two of CSI reports relate to a respective one of a plurality of component carriers, and one of the two CSI reports is a CSI report containing a Rank Indicator (RI), and one of the two CSI reports is a CSI report containing a Channel Quality Indicator (CQ), only the CSI report containing the RI on the physical uplink channel, and the baseband processor and RF transceiver further operable to receive, from the UE, when two CSI reports are to be transmitted by the UE on a physical uplink channel in a same subframe, and each of the two of CSI reports relate to a respective one of a plurality of component carriers, and each of the two CSI reports is a CSI report containing a Rank Indicator (RI), only the CSI report containing a RI associated with a highest priority component carrier.
8. The base station of claim 7, wherein the physical uplink channel is a physical uplink control channel (PUCCH).
9. The base station of claim 7, wherein the baseband processor and RF transceiver are operable to receive a CSI report from the UE every subframe.
10. The base station of claim 7, wherein the baseband processor and RF transceiver are further operable to transmit, to the UE, information to configure the UE with component carrier priority information.
11. The base station of claim 7, wherein the baseband processor and RF transceiver are further operable to transmit, to the UE, information to configure the UE to perform periodic CSI reporting associated with one or more of the plurality of component carriers.
12. The base station of claim 7, wherein the baseband processor and RF transceiver are further operable to transmit, to the UE, a medium access control (MAC) activation command to activate one of the plurality of component carriers.
13. A non-transitory computer readable storage medium comprising executable code stored thereon



that, when executed by a processor, causes a base station to: receive, from a user equipment (UE)), when two channel state information (CSI) reports are to be transmitted by the UE on a physical uplink channel in a same subframe, and each of the two of CSI reports relate to a respective one of a plurality of component carriers, and one of the two CSI reports is a CSI report containing a Rank Indicator (RI), and one of the two CSI reports is a CSI report containing a Channel Quality Indicator (CQ), only the CSI report containing the RI on the physical uplink channel; and receive, from the UE, when two CSI reports are to be transmitted by the UE on a physical uplink channel in a same subframe, and each of the two of CSI reports relate to a respective one of a plurality of component carriers, and each of the two CSI reports is a CSI report containing a Rank Indicator (RI), only the CSI report containing a RI associated with a highest priority component carrier.

**14.** The non-transitory computer readable storage medium of claim 13, wherein the physical uplink channel is a physical uplink control channel (PUCCH).

**15.** The non-transitory computer readable storage medium of claim 13, wherein a CSI report is received from the UE every subframe.

**16.** The non-transitory computer readable storage medium of claim 13, further causing the base station to: transmit, to the UE, information to configure the UE with component carrier priority information.

**17.** The non-transitory computer readable storage medium of claim 13, further causing the base station to: transmit, to the UE, information to configure the UE to perform periodic CSI reporting associated with one or more of the plurality of component carriers.

**18.** The non-transitory computer readable storage medium of claim 13, further causing the base station to: transmit, to the UE, a medium access control (MAC) activation command to activate one of the plurality of component carriers.

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